

RESEARCH ARTICLE

Structures of the co-branching reactive products of isotactic polypropylene with high-density polyethylene and the effect on the in situ compatibilization of mixed recycled materials

Yinhan Xu¹ | Jun Wang¹ | Zhu Luo¹  | Jianjun Li² | Bai Xue¹ |
Xingjiang Chen³ | Xiaolong Li⁴ | Le Yang⁵ | Changkai Linghu¹ | Yao Tao¹

¹College of Materials and Metallurgy, Guizhou University, Guiyang, China

²Kingfa Science & Technology Co., Ltd, Guangzhou, China

³Guizhou Academy of Sciences, Guiyang, China

⁴Hubei Engineering Research Center for Biomaterials and Medical Protective Materials, School of Chemistry and Chemical Engineering, Huazhong University of Science & Technology, Wuhan, China

⁵School of Materials and Energy Engineering, Guizhou Institute of Technology, Guiyang, China

Correspondence

Zhu Luo, College of Materials and Metallurgy, Guizhou University, Guiyang 550025, China.

Email: luozhu2000@sina.com

Jianjun Li, Kingfa Science & Technology Co., Ltd, Guangzhou 510665, China.

Email: 1031333109@qq.com

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Abstract

The recycling of petroleum-based polymers such as polypropylene (PP) and polyethylene (PE) is an effective means of reducing environmental pollution. However, the incompatible PP and PE blended waste is difficult to separate accurately, which reduces the quality of the recycled material. In this paper, the trifunctional acrylate monomer (trimethylolpropane triacrylate), peroxide (2,5-dimethyl-92,5-di[tert-butylperoxy] hexane) and free radical activity modifier (zinc dimethyldithiocarbamate) were applied to in situ generate the copolymer of isotactic polypropylene and high-density polyethylene (iPP-co-HDPE) with a long-chain co-branched structure via a melt blending method. The co-crystallization of the branched copolymer chain segments with the chemical identical homopolymers vastly improved the compatibility of iPP and HDPE components. This in situ co-branching compatibility strategy resulted in a significant improvement in the mechanical properties of PE and PP mixed wastes. The temperature gradient extraction method was used to separate the co-branching modified iPP/HDPE blends. The results confirm that the separated iPP-co-HDPE components have a highly co-branched structure. The significant improvement in compatibility resulted in a large increase in elongation at break and impact strength of the iPP/HDPE blends, which provides an economic and effective method for the recycling of mixed PE and PP wastes.

KEYWORDS

applications, recycling, synthesis and processing techniques

1 | INTRODUCTION

Plastics are ubiquitous in our lives and have greatly improved the quality of human life, becoming an integral part of social development and progress.¹ Among them, high-density polyethylene (HDPE) and isotactic polypropylene (iPP) are two of the most widely used thermoplastics globally, due to their excellent overall mechanical properties, processability and low cost.² However, their durability and non-degradability mean that it is difficult or impossible for these materials to be assimilated by nature.^{3,4} Cumulative and irreversible plastic pollution is a serious threat to the global environment, and it is predicted that there will be 12 million tons of waste plastic in the natural environment by 2050.^{5–7} Although most commercial plastics, such as PE and PP, are theoretically physically recyclable, many problems exist in plastic recycling, including the challenge of collecting and sorting huge amounts.^{1,8,9} Currently, only about 12% of global plastic is recycled,¹⁰ while the vast majority of the rest is discarded, incinerated or deposited in landfill.¹¹ As the main types of polyolefins, PE and PP account for approximately two-thirds of the worldwide plastics market.^{3,12} Despite having similar chemical structures, PE and PP are incompatible non-polar polyolefins. Furthermore, a large amount of mixed recovered PE and PP leads to severe delamination and performance degradation of the recycled products.^{13,14} Therefore, effectively improving the compatibility of PP and PE mixed wastes will not only help progress the recycling of commodity plastics, but also help develop a sustainable global plastics market.

Currently, the techniques of compatibilization modification of PE and PP blends can be typically divided into two main categories: non-reactive and reactive compatibilization. The non-reactive compatibilization is usually to add graft copolymers or block copolymers as compatibilizers which migrate and remain at the interface between the two phases and “emulsify” the blend to improve compatibility of the incompatible blends.^{15,16} However, the high dosage of common compatibilizers and high cost limit the application of this approach in plastics recycling. Reactive compatibilization for polymer blends is often via the in-situ formation of a compatibilizer through covalent or ionic bonding.^{17–20} Comparatively, reactive compatibilization has the advantages of convenient preparation, lower cost, more effective enrichment of the compatibilizer at the phase interface. Wang et al.²¹ synthesized a series of polyethylene-graft-atactic polypropylene (PE-g-aPP) graft copolymers through copolymerization of a cyclooctene terminated aPP macromonomer and cyclooctene using ring opening metathesis polymerization (ROMP) and subsequent hydrogenation, the PE-g-aPP graft copolymers exhibit

dramatically modified dynamic mechanical properties. Gu et al.²² reported that the compatibility of the two phases of PE and PP was improved by the addition of DCP, which enhanced the impact strength. Yao et al.²³ prepared high-density polyethylene grafted isotactic polypropylene (PP-g-HDPE) by the imidization reaction between maleic anhydride grafted polyethylene and amine-grafted polypropylene in a xylene solution. However, it is necessary to further improve the compatibilization efficiency and restrain side reactions, so as to be widely used in the cheap hybrid polyolefin recycling materials.

Currently, due to their excellent compatibilization effects, the preparation of controllable long chain segment copolymers and their structure–property relationships are becoming a hot research topic in regards to the high-quality recycling of polyolefin mixed waste plastics.^{16,24–29} In our previous study,³⁰ trimethylolpropane triacrylate (TMPTA), 2,5-dimethyl-92,5-di(tert-butylperoxy) hexane (DHPB), and zinc dimethyldithiocarbamate (ZDMC) were used to economically and conveniently generate the long chain co-branched copolymers in situ for compatibilization of HDPE and IPP blends. Consequently, superior compatibility and mechanical properties of the HDPE/PP blends were obtained. However, the application effect of the in-situ compatibilization on PP and PE mixed recycling material, structure of the reaction product and the compatibilization mechanism are still not clear. Compared with the structures of the novel compatibilizers with well-defined long chain segment copolymers, the structures of the in-situ generated copolymers is complicated. The clarification of the compatibilization mechanism based on the analysis of the in-situ product structure is very important for the implementation of this compatibilization strategy in high quality recycling of the mixed polyolefin waste.

Further study is required to enable the economical and convenient in-situ compatibilization strategy to be used in practical applications. In this study, we established the reaction system using the peroxide, trifunctional acrylate monomer, and free radical activity modifier, effectively generating the long-chain branched (LCB) copolymer of iPP and HDPE, in situ, as a compatibilizer for iPP/HDPE blends. This resulted in greatly improved blend compatibility and mechanical properties of blends of mixed PE and PP wastes. In addition, the products of in situ compatibilization were separated via a temperature gradient extraction classification method using n-octane as the solvent. Subsequently, the structure of the compatibilizer and the in-situ compatibilization mechanism was investigated. The results showed that the in-situ compatibilization method has great potential in the recycling and reusing of the mixed recycled PP/PE.

2 | EXPERIMENTAL

2.1 | Materials

Isotactic polypropylene (T30S, MFI = 2.5 g/10 min, at 190°C with 2.16 kg) and high-density polyethylene (5000S, MFI = 1.7 g/10 min, at 190°C with 2.16 kg) were both obtained from the China Petroleum Lanzhou Petrochemical Corporation; 2,5-dimethyl-2,5-di(tert-butylperoxy) hexane (DHBP) and trimethylolpropane triacrylate (TMPTA) were provided by the Aladdin Reagent Company. Zinc dimethyldithiocarbamate (ZDMC) was purchased from Hebi Lianhao Chemical Company. Recycled PP material was from injection parts and recycled PE material was from plastic bottles, both provided by China Guizhou Kekot materials company.

2.2 | Sample preparation

The DHBP and TMPTA were dissolved in an appropriate amount of acetone and sprayed evenly onto the iPP and HDPE mixture with different proportion, the ZDMC was thoroughly mixed and then placed in a ventilator for 24 h to allow the acetone to completely evaporate. The blends were subsequently mixed in a torque rheometer with a rotor speed of 85 rpm at 200°C for 240 s and finally, the modified iPP/HDPE blended specimens were produced. As a control for the experiment, the same reactive processing system and method were used for pure PP and pure PE to prepare LCB PP and LCB PE, which were then blended in a particular ratio to form LCB iPP/HDPE. Figure 1 displays the schematic diagram of preparing the modified material and the effect of improving mechanical properties, besides, the formulation of modified iPP/HDPE blends are summarized in

Table 1 (the contents of the additives was determined by orthogonal experiments in our previous research).

2.3 | Purification and separation of modified samples

In order to further analyze the structure of the iPP/HDPE modified specimens we purified the prepared specimens to remove unreacted additives. The modified iPP/HDPE (50/50) blend was completely dissolved in xylene at 140°C. After cooling to room temperature, methanol was added to precipitate the modified sample, which was then filtered three times to remove any unreacted small molecular substances. The purified sample was then dried in a vacuum oven at 60°C for 12 h, and stored in a desiccator for subsequent testing and characterization.

The iPP-co-HDPE for in situ compatibilization is formed in the process of melt reaction, therefore, it is necessary to separate the modified iPP/HDPE blends to obtain iPP-co-HDPE in order to characterize its structure and properties. The modified iPP/HDPE (50/50) blend were separated using temperature gradient extraction fractionation^{31,32} using *n*-octane as a solvent. First, 10 g of the purified sample was placed in *n*-octane at 125°C for 72 h, the solution was then cooled to room temperature and separated by filtration to obtain a viscous paste. The paste was extracted in *n*-octane at 95°C for 72 h and the filtrate was collected. The filtrate solution was then concentrated and dried by rotary evaporator, and the obtained solid was named G95 (HDPE). The remaining sample was then stored at 110°C for 72 h. iPP-co-HDPE (named G110) was separated from the remaining sample by following the same procedure. Finally, iPP, the residual paste after extraction was obtained (named G125).

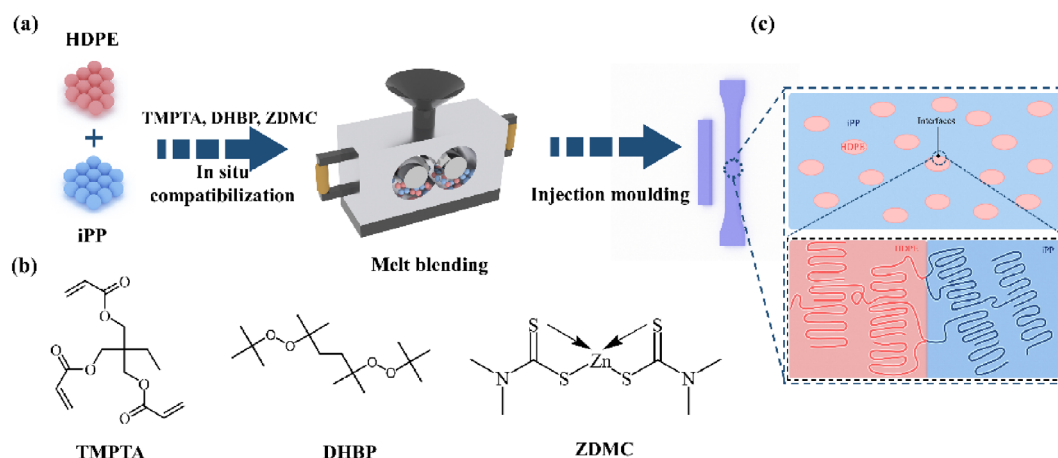


FIGURE 1 (a) Schematic representation of modified iPP/HDPE blends. (b) Chemical structure of TMPTA, DHBP, and ZDMC. (c) Schematic of the two-phase interface [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]

TABLE 1 Formulation of modified iPP/HDPE blends

	iPP (wt%)	HDPE (wt%)	DHBP (wt%)	TMPTA (wt%)	ZDMC (wt%)
iPP	100	0	0.05	1.2	0.1
HDPE	0	100	0.05	1.2	0.1
unmodified iPP/HDPE (70/30)	70	30	0	0	0
modified iPP/HDPE (70/30)	70	30	0.05	1.2	0.1
unmodified iPP/HDPE (50/50)	50	50	0	0	0
modified iPP/HDPE (50/50)	50	50	0.05	1.2	0.1
recycled PP/PE (70/30)	70	30	0.05	1.2	0.1
recycled PP/PE (30/70)	30	70	0.05	1.2	0.1

2.4 | Characterization and measurements

2.4.1 | Mechanical properties tests

The tensile strength test was performed on a universal testing instrument (CMT6104, MTS System Corporation) with a crosshead speed of 50 mm/min at 20°C, according to ISO standard 1184-198. At least five bars were evaluated for each sample.

Impact testing was performed on a ZBC1400-B (Meters Industrial Systems China Company) cantilever beam impact tester with a pendulum energy of 2.75 J at a test temperature of 20°C. Here, five impact test specimens were measured to attain the average. All samples were prepared by a PL860 injection molding machine (Wuxi Haitian Machinery company, China) at a temperature of 200°C.

2.4.2 | Scanning electron microscopy (SEM)

The specimen fracture morphology was investigated via a Japanese JEOL JSM-7500F scanning electron microscopy (SEM) with an accelerating voltage of 5 kV. The specimens were etched in a mixture of potassium permanganate and concentrated sulfuric acid for 48 h at room temperature and then sputtered with a thin gold layer.

2.4.3 | Successive self-nucleation and annealing (SSA) experiments

Successive self-nucleation and annealing (SSA) experiments³³ were performed on a Q10 differential scanning calorimeter (DSC, TA, USA) to effectively amplify the crystalline structural differences of the iPP/HDPE blends. A schematic representation of the SSA experiment is shown in Figure 2. As shown, during the first step the

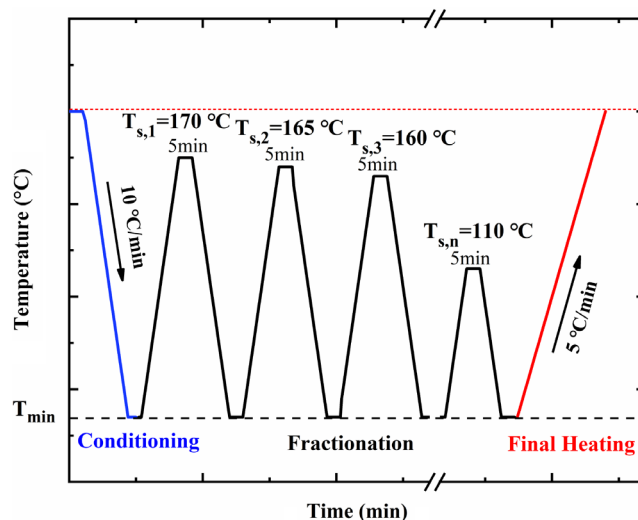


FIGURE 2 Temperature scan procedure for SSA thermal treatment, the T_s temperatures varied from $T_{s,1}$ to $T_{s,n}$, onset of each sample at 5°C intervals. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

SSA protocol rapidly increased in temperature to 200°C and is maintained for 5 min. It was then cooled to 30°C at 10°C/min, in order to create an initial “standard” thermal history. Subsequently, a heating scan at 10°C/min was performed up to a selected first self-nucleation and annealing temperature ($T_{s,1}$), where the sample was isothermally maintained for 5 min before cooling again at 10°C/min to 30°C. During the repeated heating-cooling process, T_s was lowered by 5°C intervals with respect to the previous step, within the range of 170–110°C. Finally, the sample was heated to 200°C at a rate of 5°C/min to record the final SSA heating curve.

2.4.4 | Nuclear magnetic resonance (NMR)

¹H NMR sample (20 mg of each sample) was dissolved in deuterated o-Dichlorobenzene at 125°C and agitated until

a homogeneous solution was reached. All ^1H NMR spectra were acquired via a Bruker AVANCE NEO 600 MHz spectrometer. The ^1H NMR experiments were performed utilizing a 10 s relaxation delay and an acquisition interval time of 5 s. A total of 512 transient scans were collected.

2.4.5 | Raman spectroscopy

The samples were examined by a laser confocal micro Raman spectrometer DXR 2 by Thermo Fisher, USA, with a detection time of 3 s and 30 exposures. The spectral acquisition wavenumber was in the range of $50\text{--}3367\text{ cm}^{-1}$ with a maximum energy of 10 mW and laser wavelength of 532 nm.

2.4.6 | Rheological measurement

Dynamic frequency sweeping (DFS) measurements were performed using a strain-controlled ARES-G2 rheometer (TA Instruments) with a flat plate mode, for all the blends. The parallel plate diameter was 25 mm and the plate spacing was set at 1 mm. Primarily, a dynamic strain sweep was performed to determine the linear viscoelastic range of the specimen at a test temperature of 200°C . The result revealed a strain of 1%. Subsequently, a dynamic frequency sweep was performed in the linear viscoelastic region in a sweep frequency range of $0.1\text{--}500\text{ rad/s}$ at a constant test temperature of 200°C .

2.4.7 | Crystallization behaviors

Differential scanning calorimetry (DSC, TA Q10) was utilized to study the crystallizing and melting behavior of the iPP/HDPE blends. 6–10 mg of samples were sealed in an aluminum pan and loaded in a DSC instrument with a N_2 flow rate of 40 ml/min. The samples were rapidly heated to 200°C and maintained at this temperature for 5 min to erase the previous thermal history. They were then cooled to 40°C at 5°C/min , before being reheated to 200°C at 5°C/min . The crystallization and melting curves were recorded and the crystallization temperature (T_c) and melting temperatures (T_m) were obtained from the second heating cycle.³⁴

3 | RESULTS AND DISCUSSION

3.1 | Mechanical properties of blends

The mechanical property test is a critical means to verify the compatibilization effect of the in situ co-branched

reactive blending system. The blend of LCB PP and LCB PE was introduced as a contrast to evaluate the effect of LCB co-branched polymer on the compatibility of iPP/HDPE blends. The same system and method were used to prepare LCB PP and LCB PE, which were then blended in 50/50 ratio to form LCB iPP/HDPE. The mechanical properties of iPP, HDPE, and the different proportions of iPP/HDPE blends are shown in Figure 3. The HDPE and iPP homopolymers display good deformability, and the elongation at break of them were 621.1% and 680.3%, respectively. However, the break at elongations of the unmodified iPP/HDPE (70/30) and (50/50) blends decrease to 75.2% and 123.3%, while the elongation at break of the modified iPP/HDPE (70/30) blend is 475.8%, and the elongation at break of the modified iPP/HDPE (50/50) blend reaches 522.2%. Contrastingly, the break at elongation of LCB iPP/HDPE (50/50) blend is only 162.3%. Tensile strength of the modified iPP/HDPE blends increases slightly compared to the unmodified blends. The notched impact strengths of unmodified iPP/HDPE (50/50) and unmodified iPP/HDPE (70/30) blends are 6.1 and 4.8 kJ/m^2 , respectively, and the notched impact strength of the LCB iPP/HDPE (50/50) blend increases to 7.28 kJ/m^2 , while the notched impact strengths of the modified iPP/HDPE (50/50) and modified iPP/HDPE (70/30) blends increase to 23.1 and 9.8 kJ/m^2 , respectively. This shows that the LCB iPP-co-HDPE compatibilization effect is much superior to that of LCB iPP and/or LCB HDPE on the iPP/HDPE blends, indicating that the compatibilization effect results from the co-branched reaction products rather than the single component branched reaction products. The impact strength of the modified iPP/HDPE (70/30) blend is also much higher than that of the pure blend, with the value being increased by more than double. Specifically, the impact strength of the modified iPP/HDPE (50/50) blend is similar to that of pure HDPE and about four times that of the pure blend, displaying a superior toughening effect to the modified iPP/HDPE (70/30) blend. This is due to the fact that the iPP with low impact toughness is the major phase in the iPP/HDPE (70/30) blend.^{15,35,36}

It is very important to confirm the usefulness of the in-situ compatibilization for the recovery of hybrid polyolefin. The mixed PP and PE recycled materials purchased from the market are modified with the same in-situ compatibilization method in PP/PE 7:3 ratio and 3:7 ratio. As shown in Figure 4a, the simply blended specimens suffer severe delamination and poor compatibility. However, the surface of the co-branched modified specimens was smooth and flat, and displayed a good in-situ compatibilization effect on the mixed recycled materials (Figure 4b). In addition, the elongation at break (Figure 4c) and notched impact strength (Figure 4d) are significantly improved after the co-branching modification,

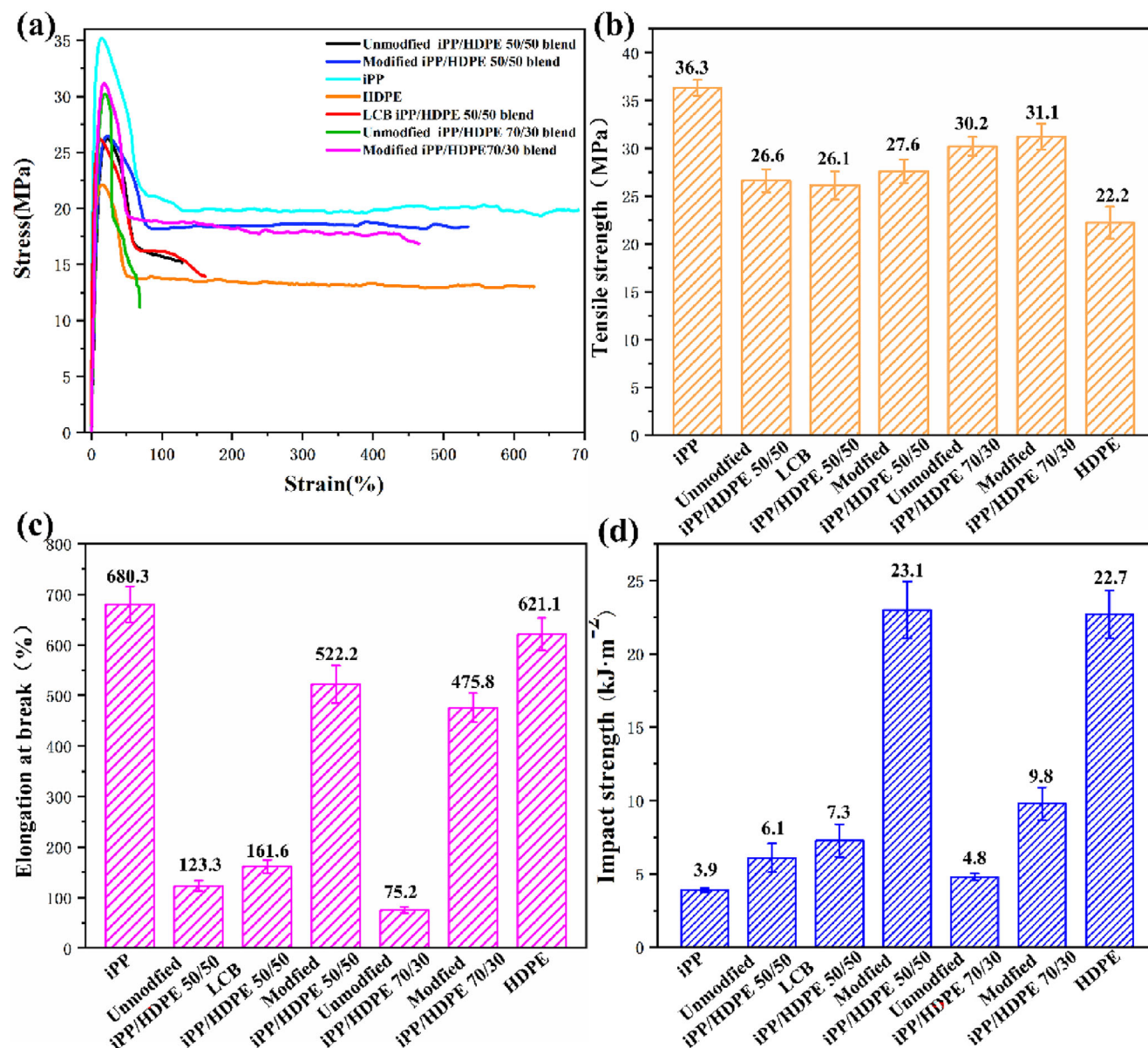


FIGURE 3 The mechanical properties of iPP, HDPE, and the different proportions of iPP/HDPE blends. (a) Stress strain curves. (b) Tensile strength. (c) Elongation at break. (d) Notched impact strength [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

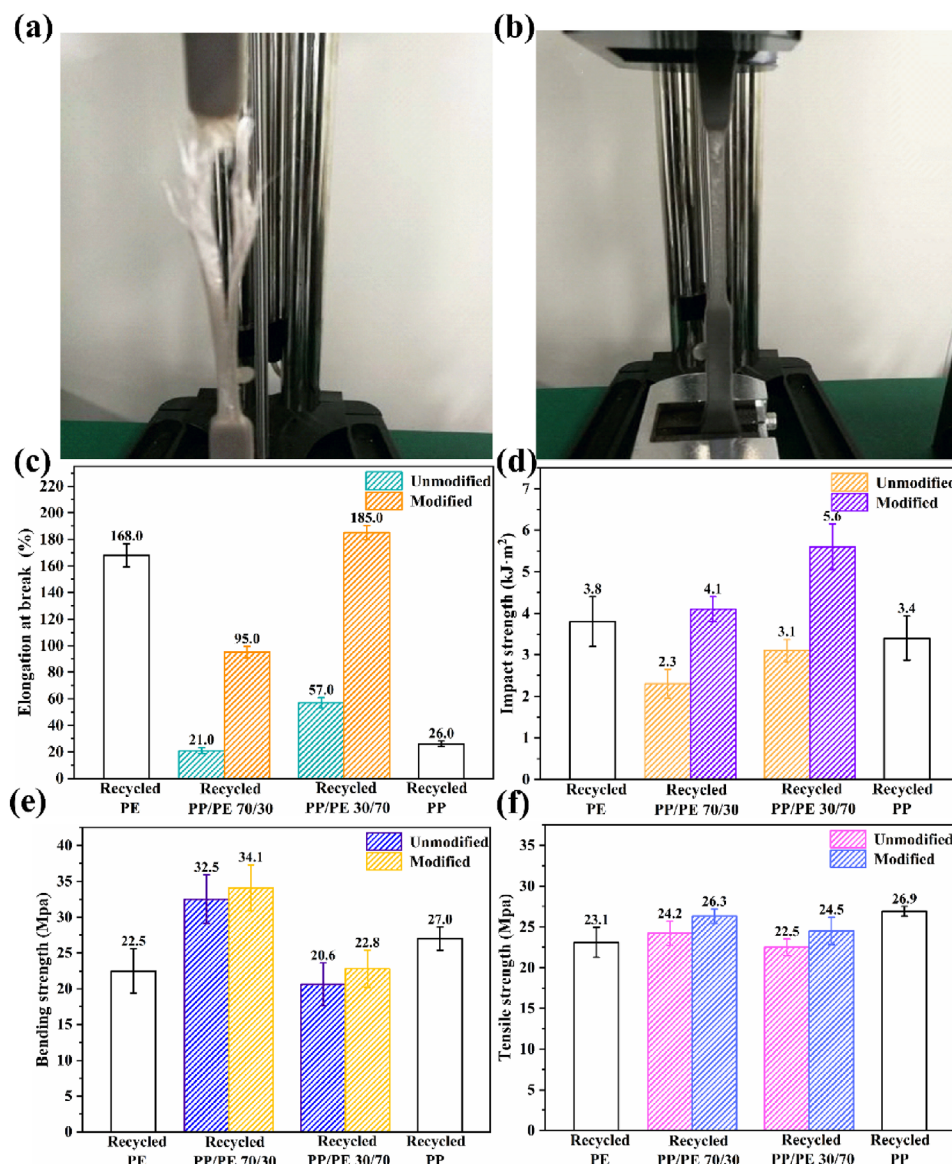
which is closely related to the compatibility. The tensile and bending strengths of the modified iPP/HDPE blends increase slightly compared with that of the unmodified blends. Compared with the unmodified recycled PP/PE (70/30) blend, the elongation at break of the co-branched modified recycled PP/PE (70/30) blend is 95%, which is nearly 4.5 times higher than that of the unmodified recycled blend, the elongation at break of recycled PP/PE (30/70) blend increased from 57% of the unmodified specimen to 185% of the modified specimen. In addition, the notched impact strength of the unmodified recycled PP/PE (70/30) and co-branched modified recycled PP/PE (70/30) blends are 2.3 and 4.1 kJ/m², respectively. The notched impact strength of the recycled PP/PE (30/70) blend

increased from 3.1 of the unmodified specimens to 5.6 kJ/m² of the modified specimen. The mechanical properties of the modified recycled blends by co-branching are obviously improved, indicating that the in-situ co-branching compatibilization has applicability and effectiveness for improving the performance of the mixed recycled polyolefin.

3.2 | Nuclear magnetic resonance analysis

Based on the characterization of material properties, we predict the melt co-branching compatibilization reaction

FIGURE 4 (a) Tensile specimens of recycled material with PP/PE (70/30) unmodified blends. (b) Tensile specimens of recycled material with PP/PE (70/30) modified blends. (c) Elongation at break. (d) Notched impact strength. (e) Bending strength. (f) Tensile strength [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]



process (Figure 5a). In the process of in situ generating the copolymer of isotactic polypropylene and high-density polyethylene (iPP-co-HDPE) with a long-chain co-branched structure via a melt blending method, the hydrogen atoms on the HDPE molecular chain methylene group and on the PP molecular chain tertiary carbon atoms are easily seized by primary radicals to form macromolecular radicals. Meanwhile the carbon-carbon double bonds of the polyfunctional acrylate monomer (M) are opened and combine with the macromolecular radicals to form a branched structure centered on polyfunctional chain segments.³⁰ The in situ co-branching product was verified by ¹H NMR (Figure 5b,c). Prior to the ¹H NMR characterization of modified iPP/HDPE 50/50 blends, the modified samples were purified to remove impurities such as unreacted TMPTA. The ¹H

NMR spectra of the unmodified blended specimen and the modified specimen are shown in Figure 5b,c. Compared with the unmodified blended specimens, new characteristic peaks appeared at $\delta = 0.46$ ppm (a) and $\delta = 1.52$ ppm (b) in the modified iPP/HDPE 50/50 blend ¹H NMR spectra, representing the terminal $-\text{CH}_3$ and internal $-\text{CH}_2$ groups present in TMPTA, respectively. The $-\text{CH}_2$ group protons attached to the carbonyl functional group in TMPTA appeared at $\delta = 2.45$ ppm (d). The peak at $\delta = 2.39$ ppm (c) corresponded to the $-\text{CH}$ proton of the ethylene chain linked to the TMPTA.^{37,38} The peak at $\delta = 2.57$ ppm (e) indicated the $-\text{CH}$ proton from the propylene chain bonded to TMPTA. This confirms the successful synthesis of the TMPTA-centered branched iPP-co-HDPE by the linkage of HDPE and iPP to TMPTA.

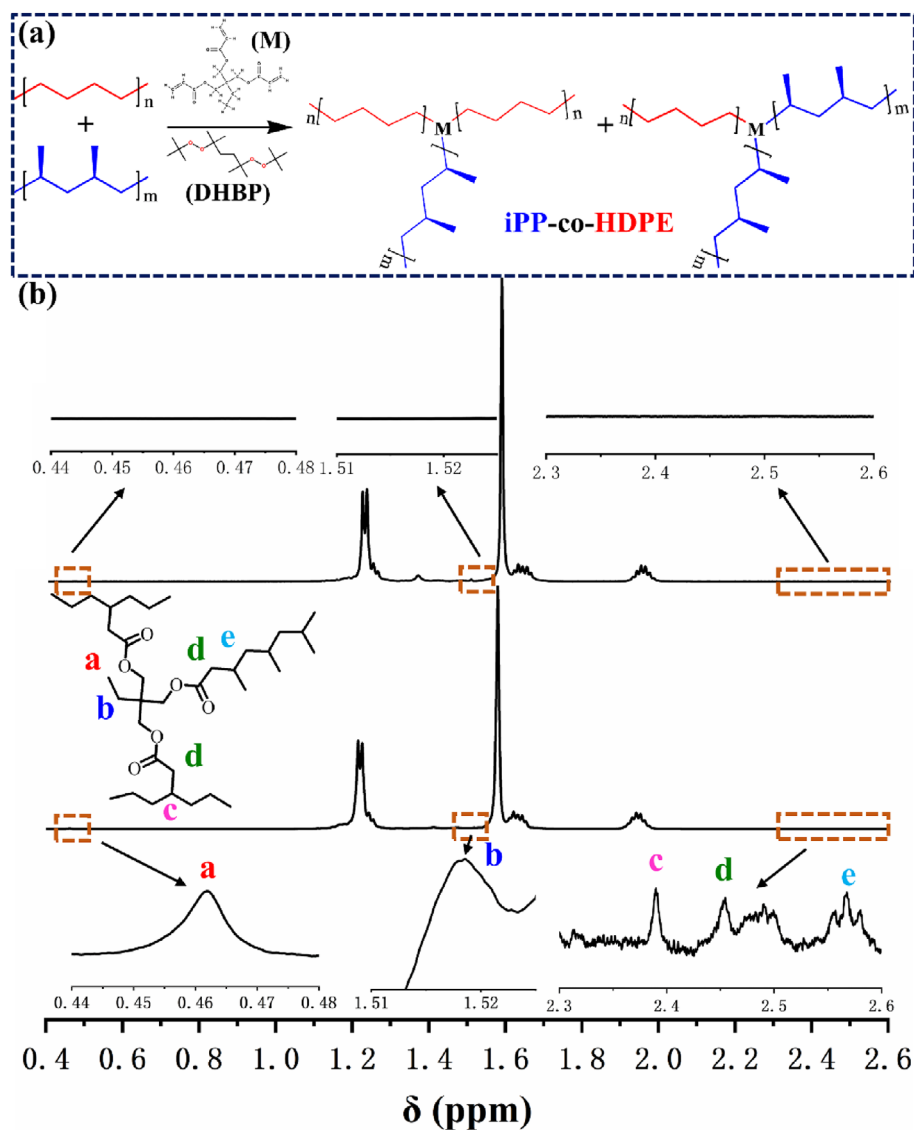


FIGURE 5 (a) Synthesis scheme of long-chain branched iPP-co-HDPE copolymer. ^1H NMR images of (b) unmodified blended and (c) modified specimens. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]

3.3 | Morphology analysis of blends

The isothermally (130°C) crystallized specimens and the injection molded specimens of unmodified and modified iPP/HDPE (70/30) blend are etched with $\text{KMnO}_4\text{-H}_2\text{SO}_4$ solution. Since iPP and HDPE components are incompatible with each other, phase separation can easily occur. Furthermore, HDPE possesses a stronger resistance to acid etching than iPP under the same conditions. Figure 6a,b show SEM images of the samples isothermally crystallized at 130°C . The etched morphology of the modified iPP/HDPE (70/30) blend displays that the trenches are clearly centered at the two-phase interface and radially distributed in the PE phase (Figures 6b), which indicates the PP crystal nucleated at the interface grows unto the PE enriched phase region. The PP crystalline region is etched away to form trenches centered near the two-phase interface, and the PE next to the trench

remained. As the LCB copolymer is distributed at the interface of the two phases and its PP chain segments can be crystallized at a higher temperature, the PP crystal extends into the HDPE-rich phase region and the lamellar PP crystals are etched away, leaving obvious grooves (Figure 6b). As a contrast, the unmodified copolymer morphology do not display this feature (Figure 6a). Obviously, iPP homopolymer is co-crystallized with the PP branched chain segments of the in situ generated LCB co-polymer which distributes in the PE-rich phase region near the interface.

Figure 6c,d shows the etched surface morphology of the injection molded unmodified and modified samples, respectively. Comparing the morphology of injection modified blends with that of injection unmodified blends, the spherical crystals of the unmodified blend were large and there is an obvious phase interface, however, for the modified blend, the interface between the two phases

became indistinguishable and the spherical crystals are much smaller. This is due to the significant improvement of compatibility between HDPE and iPP, which cannot be achieved by a simple physical entanglement, indicating that the co-crystalline structure can be also formed at

the interface under the injection molding condition to connect the two phases, and the grain size is reduced by nucleation of the long-branched chains. The co-crystallization leads to an increase in the compatibility of the iPP/HDPE blends, which further improves its mechanical properties.

Figure 6e shows the compatibilization mechanism of the iPP/HDPE long-branched copolymers. Long-chain branched copolymers are enriched at the interface, and the copolymer chain segments form a co-crystalline structure with the chemically identical homopolymers, facilitating efficient compatibilization. The effective stitching between iPP and HDPE phases reduces the interfacial tension and improves the compatibility of the two phases, thus improving the mechanical properties of the iPP/HDPE blends.^{38,39}

3.4 | Crystallization behavior of blends and the separated components

The successive self-nucleation/annealing (SSA) technique is a useful tool to effectively amplify the differences in crystalline structure.⁴⁰ The SSA heating curves of the iPP/HDPE (50/50) and (70/30) blends are displayed in Figure 7a,b. The in-situ generation of LCB structures in modified blends led to a degree of increase in the melting temperature for each SSA thermal classification, suggesting that the nucleation of long-chain branched copolymers promotes the perfection of a part crystal structure during the thermal classification process. With the decrease of temperature during cooling crystallization, the mobility of HDPE molecular chains decreases, and

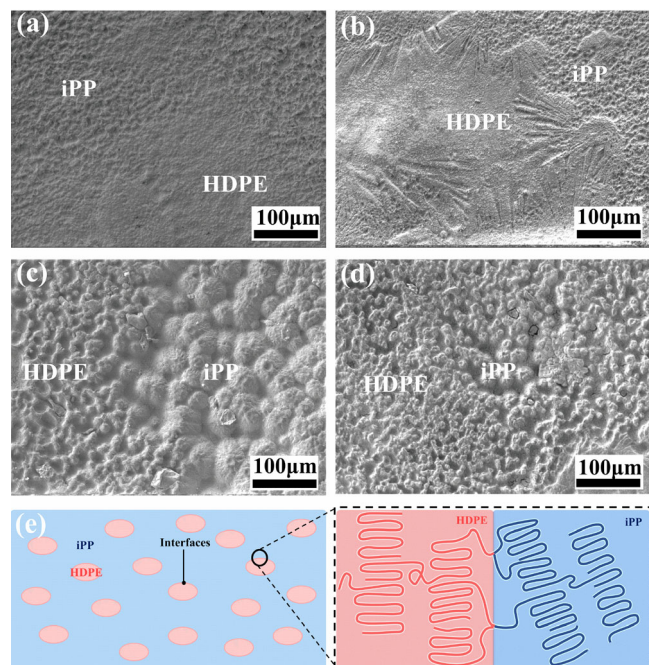


FIGURE 6 The SEM micrographs of (a) unmodified and (b) co-branched iPP/HDPE modified (70/30) blends. The injection molded specimens of (c) unmodified and (d) co-branched modified iPP/HDPE (70/30) blends. (e) 2D schematic illustration of the two-phase interface [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]

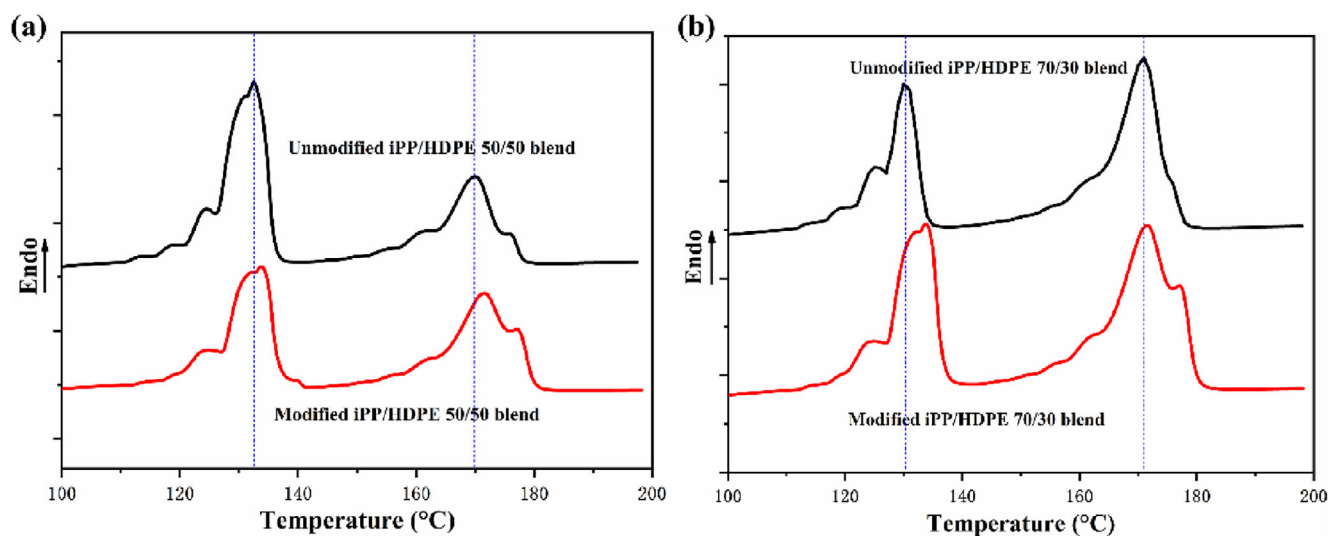


FIGURE 7 Temperature rise curves of iPP/HDPE blends after thermal classification. (a) iPP/HDPE 50/50 blends. (b) iPP/HDPE 70/30 blends [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]

TABLE 2 Crystallinity and crystallization onset temperature of different samples

Sample	Crystallinity (%)		Crystallization onset temperature (°C)	
	iPP	HDPE	iPP	HDPE
HDPE		59.39		120.28
iPP	45.63		124.63	
Unmodified iPP/HDPE (50/50)	29.10	36.63	125.39	119.34
Modified iPP/HDPE (50/50)	39.61	47.49	133.60	121.41

the perfection of crystallization reduces, resulting in a part of crystal structure molten at lower temperature in the heating curve in Figure 7a,b. For the unmodified blended sample, PE crystallization near the interface is constrained by the iPP phase at the end of crystallization, which aggravates the reduction in crystallization perfection and produces more low temperature molten crystal compared to the modified iPP/HDPE blends. The LCB iPP-co-HDPE generated in situ during co-branched reaction is enriched at the phase interface, and its ethylene branched chains are co-crystallized with the homopolymer HDPE. This reduces the iPP constraint on crystallization of HDPE and improves the crystalline perfection.

It can be seen from Table 2 that the crystallinity of pure iPP and HDPE is 45.63% and 59.39%, respectively. In contrast, the iPP and HDPE components in the unmodified blends crystallized in their respective cells, therefore, the iPP and HDPE components in the molten blends repelled each other and formed incompatible separated phase structures during the crystallization process, and this mechanism resulted in a lower crystallinity of iPP and HDPE in the pure blends than that of the single components, 29.10% and 36.63%, respectively. For the modified iPP/HDPE (50/50) blend, the long-branched LCB copolymer chains are readily entered into the lattice and play a crystalline nucleation role, resulting in improved crystallinity compared to the unmodified iPP/HDPE (50/50) blend.

Figure 8a,b show the DSC cooling crystallization curves and melting curves of iPP, HDPE, and iPP/HDPE (50/50) blend (unmodified and modified). The crystallization curves in Figure 8a clearly show that the iPP crystallization peaks in the modified iPP/HDPE (50/50) blend shifted to the higher temperature of 133.6°C (Table 2), compared with the unmodified iPP/HDPE (50/50) blend and pure iPP. The long-branched LCB copolymer chains are readily entered into the lattice and play a crystalline nucleation role, which raises the starting crystallization temperature of the modified iPP/HDPE blends. In Figure 8b, the melting temperature of iPP in modified iPP/HDPE (50/50) blend is slightly higher than that in unmodified iPP/HDPE (50/50) blend and pure iPP, due to the enhanced perfection of the part crystal structure crystallized at a higher temperature.

Figure 8c,d shows the DSC analyses of the components separated from the modified iPP/HDPE (50/50) blend by extracting. Compared with G95 (HDPE) and G125 (iPP), the G110 component (iPP-co-HDPE) had distinct melting and crystallization peaks attributed to PP and PE. In addition, its polypropylene chain segment crystallized at a significantly higher temperature and the polyethylene chain segment crystallized at a lower rate. Some studies have shown that the crystallization rate of simple PE-PP blends is not dissimilar to that of single-component homopolymers.⁴¹ The crystallization rate of PE in G110 decreased due to the interference of PP branched chains with the post-crystalline PE chains.

Raman spectroscopy is highly sensitive to changes in polymer microstructure and chain conformation, which is useful in the research of molecular melting and crystallization mechanisms.⁴² The Raman spectra of the three components (G95, G110, and G125) graded by temperature gradient extraction, in the wavenumber range of 500–2000 cm⁻¹, are presented in Figure 9, and the phase assignments are shown in Table 3. From Figure 9, it can be seen that the Raman spectra of G95 and G125 separated by temperature gradient extraction, only had distinct HDPE and iPP characteristic peaks respectively. Both HDPE and iPP characteristic peaks appeared in the iPP-co-HDPE (G110) Raman spectra, but the intensity of the corresponding peaks was significantly reduced.^{43,44} Furthermore, the Raman spectra of G110 at 808, 840, 1297, and 1127 cm⁻¹ showed a significant decrease in intensity due to the interference of the iPP-co-HDPE branched chains,⁴⁵ which leads to the decrease in G110 crystallinity. Meanwhile, there was no characteristic HDPE peak in the G95 Raman spectrum, and no obvious iPP characteristic peak in the G125 Raman spectrum. This indicates the high purity of the components separated by temperature gradient extraction classification.

3.5 | Dynamic shear rheology analyses of the separated components

The molecular topology structure is changed by the generation of co-branched structures. Rheological behavior

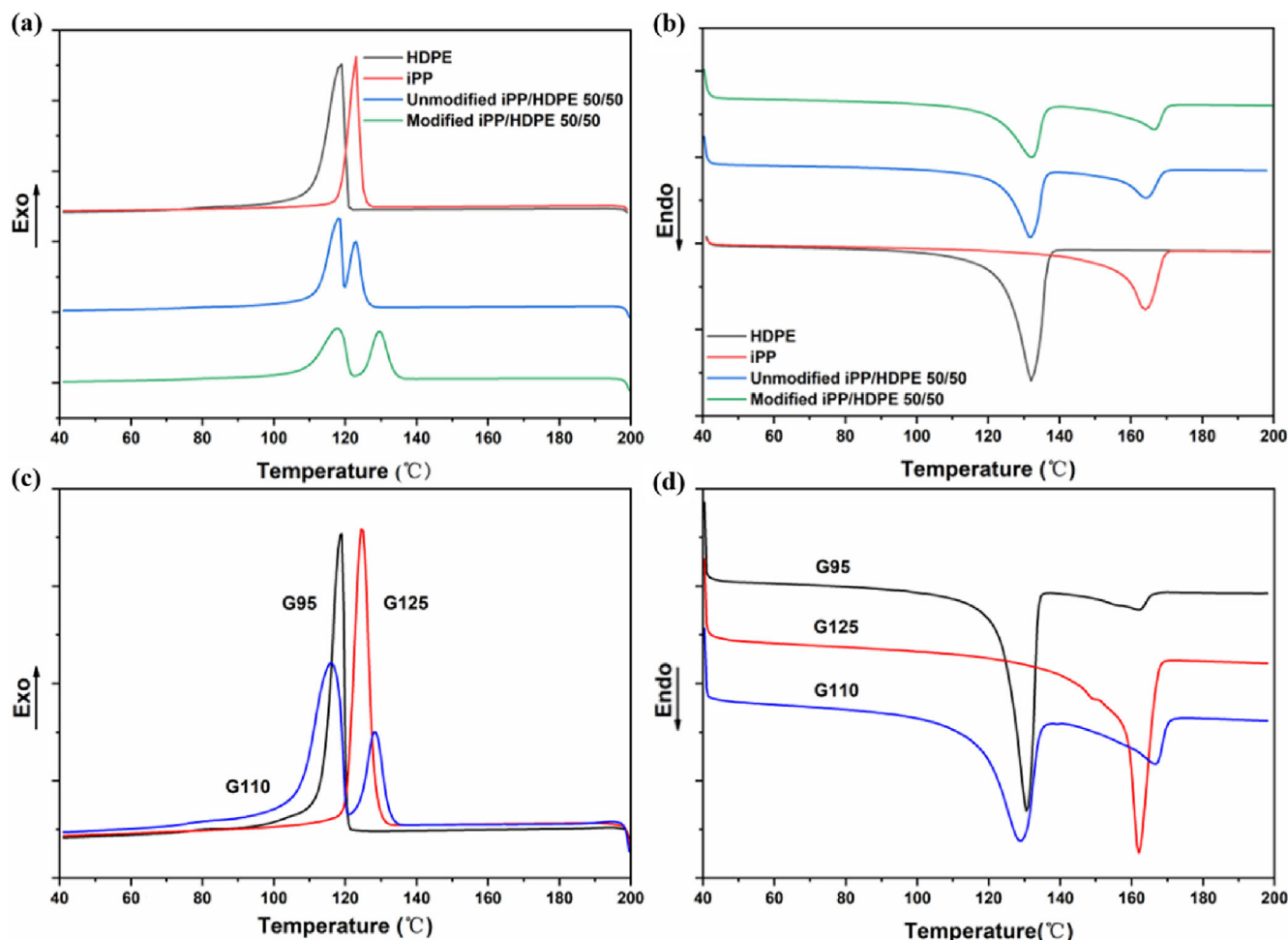


FIGURE 8 The DSC cooling curves (a) and (c). The DSC melting curves (b) and (d) [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]

is very sensitive to the molecular topology,^{48–50} therefore, it is necessary to analyze the rheological behavior of the classification components to investigate the branched reaction products in the different components.

Figure 10a shows the double-logarithmic plots of storage modulus versus angular frequency for the three components (G95, G110, and G125) graded by temperature gradient extraction, pure HDPE, and pure iPP. It can be clearly seen that iPP and HDPE exhibited the typical terminating rheological behavior of linear polymers. G95, which was HDPE, and G125, which was iPP, showed a small increase of G' in the low-frequency region, when compared with the pure samples. This is because a small amount of LCB iPP and LCB HDPE are produced during the free radical reaction, in addition to the LCB iPP-co-HDPE. Furthermore, it was displayed that the G' of the G110 component was clearly aggrandized in the low frequency region, and exhibited the obvious terminal behavior of a non-linear polymer, indicating that the main composition of G110 was a long-chain branched

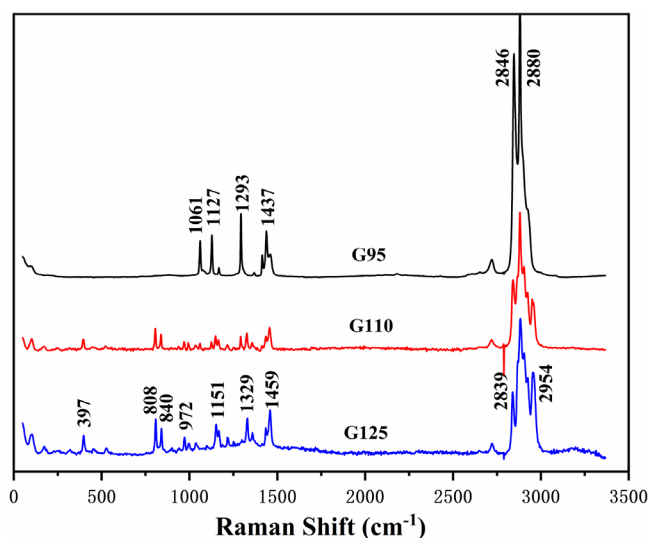


FIGURE 9 Raman spectra of HDPE (G95), iPP(G125), and iPP-co-HDPE(G110). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53170)]

copolymer generated during reactive blending. The effect of the LCB structure on the zero-shear viscosity of the branched polymer is more pronounced than the variation of G' in the low frequency region.^{51,52} Figure 10b displays the trend plots of the complex viscosity (η^*) with angular

TABLE 3 Vibrational and phase assignment of the Raman Spectrum of HDPE (G95), iPP (G125), and iPP-co-HDPE(G110)^{45–47}

Raman shift (cm ⁻¹)	Modes	Phases
iPP		
808	CH ₂ rocking, C—CH ₃ stretching, C—C stretching	Helical chain
840	CH ₂ rocking, C—CH ₃ stretching	Helical chain (12 monomer unit)
972	CH bending, C—C stretching	Helical chain (5 monomer unit)
1151	C—C stretching	
1329	CH bending, CH ₂ twisting	
1459	CH ₃ bending, CH ₂ bending	
HDPE		
1061	Asymmetric C—C stretching	Helical chain (6–8 monomer unit)
1127	C—C stretching	Helical chain (18 monomer unit)
1293	CH ₂ twisting	Helical chain (18 monomer unit)

frequency (ω) for HDPE, iPP, G95, G110, and G125. G110 exhibited continuous shear thinning behavior over the entire test frequency range resulting from the entanglement of the long-branched chains, and its η^* did not exhibit a flattening trend in the low-frequency region, while the change of η^* of linear iPP and HDPE showed a significant slowdown in the low-frequency region. Compared with HDPE and iPP, the change of η^* of G95 and G125 did not show a significant difference within the frequency range of the test. The above experimental results show that the separated iPP-co-HDPE component is the main reactive product of the branching reaction, while the contents of the generated LCBPP and LCBPE are relatively low.

4 | CONCLUSIONS

A co-branching reaction system was established by using the TMPTA, DHBP, and ZDMC to generate in situ LCB copolymers of iPP and HDPE by melt blending. The LCB copolymer chain segments co-crystallize with the chemically identical homopolymer at the interface in iPP/HDPE blends, and the compatibility of the co-branching modified blends was vastly improved. Thus, the enhanced compatibility greatly improves the mechanical properties of the co-branching modified iPP/HDPE and the mixed recycled PP and PE blends. For the co-branching modified iPP/HDPE 50/50 blend, the elongation at the break increased from 123% to 522%, and the notched impact strength increased from 6.1 to 23.1 kJ/m². For the co-branching modified mixed PP and PE recycled materials in a 7:3 ratio, the elongation at the

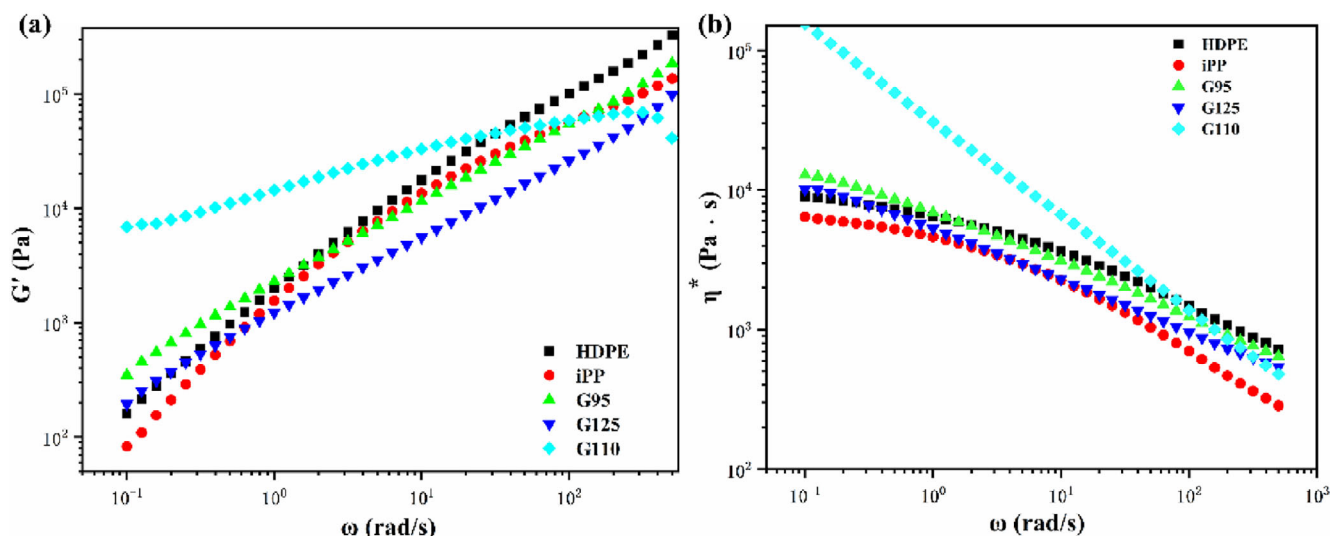


FIGURE 10 Double logarithmic plots of the angular frequencies of the storage modulus and incident complex viscosity of HDPE, iPP, G95, G125, and G110. (a) Storage modulus. (b) Complex viscosity [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

break increased from 21% to 91%, and the notched impact strength increased from 2.3 to 4.1 kJ/m², meanwhile in 3:7 ratio co-branched modified PP and PE recycled material elongation at break increased from 57% to 185% and notched impact strength increased from 3.1 to 5.6 kJ/m². In particular, the serious delamination was eliminated.

The in situ generated TMPTA-centered co-branched copolymer of iPP and HDPE was confirmed via structural analysis. The reactive product structures were further investigated by separating the components of the modified blend by using temperature gradient extraction fractionalization. The separated iPP-co-HDPE component with a highly branched LCB structure is the main reactive product of the co-branching reaction, while the contents of the generated LCB PP and LCB PE are relatively low. Additionally, iPP-co-HDPE plays a crystalline nucleation role, resulting in a higher starting crystallization temperature of the modified iPP/HDPE blends. Meanwhile, the iPP-co-HDPE enriching at the phase interface, improved the perfection of HDPE crystals. This study proposes an economical and novel strategy for the in situ compatibilization of PP/PE blends with a good application prospect in recycling waste PE and PP blends.

AUTHOR CONTRIBUTIONS

Yinhan Xu: Writing – original draft. **Zhu Luo:** Writing – review and editing. **Jun Wang:** Writing – review and editing. **Bai Xue:** Writing – review and editing. **Xingjiang Chen:** Writing – review and editing. **Xiaolong Li:** Writing – review and editing. **Le Yang:** Writing – review and editing. **Changkai Linghu:** Writing – review and editing. **Yao Tao:** Writing – review and editing.

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Zhu Luo  <https://orcid.org/0000-0001-8848-9354>

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